SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems(BL1, BL2, BL3)*		
Student Name:	Maddipati Teja	**Reg. No.:**	192425323
Section / Batch:	DAA, 2024.	**Date:**	18-2-2026

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problemanalysis	Q1 Q2,Q3	40 60 (2X30)
		GRAND TOTAL:	**100 Marks**

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I M-Teja *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M-Teja

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______	Signature: ______	Signature: ______
Date: ______	Date: ______	Date: ______

Selection Sort - Analytical problem

Initial

Array = [64, 25, 12, 22, 11]

pass 1 = [11, 25, 12, 22, 64]

pass 2 = [11, 12, 25, 22, 64]

pass 3 = [11, 12, 22, 25, 64]

pass 4 = [11, 12, 22, 25, 64]

Total comparisons

pass	comparisons
1	4
2	3
3	2
4	1
Total	10

Time complexity

case	complexity
Best	$O(n^2)$
Average	$O(n^2)$
worst	$O(n^2)$

Final sorted output

{11, 12, 22, 25, 64}

Q2

Closest pair - Brute Force

A

point

(2,3), (12,30), (40,50), (5,1), (12,

Distance formula

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Distance calculation

pair	Distance
(2,3)-(12,30)	28.79
(2,3)-(40,50)	60.44
(2,3)-(5,1)	8.61
(2,3)-(12,10)	12.21
(2,3)-(3,4)	1.41
(12,30)-(40,50)	34.41
(12,30)-(12,10)	29.83

closest pair

(2,3) and (3,4)

Distance = 1.41

Total comparisons

For 6 points, $\frac{6(6-1)}{2} = 15$

Time complexity

Best = O(n2)

Average = O(n2)

Worst = O(n2)

Convex Hull - Brute Force Approach

Points

(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)

Hull Edges

(0,0) → (3,0)

(3,0) → (3,3)

(3,3) → (0,3)

(0,3) → (0,0)

Convex Hull points

(0,0), (3,0), (3,3), (0,3)

Comparisons

No. of point pairs

$\frac{7 \times 6}{2} = 21$

Total Checks

$21 \times 5 = 105$

Time complexity

$O(n^3)$

Brute force convex hull guarantees correctness but becomes very slow for large datasets. Efficient algorithms like Graham Scan are preferred for practical application.

Q4 Searching technique - comparative Analysis

linear search:

works by checking each element sequentially.

Array
{20, 15, 35, 40, 10}

Search = 40

comparison

20

15

35

40

Total = 4 comparisons.

Binary search

Sorted Array
{10, 15, 20, 35, 40}

Search = 40

total = 3 comparisons.

Complexity

Case	Linear	Binary
Best	O(1)	O(1)
Average	O(n)	O(logn)
Worst	O(n)	O(log n)

Comparison

	Linear Search	Binary Search
Best	O(1)	O(1)
Average	O(n)	O(log n)
Worst	O(n)	O(logn)

Observation

Binary search is much faster than linear search on sorted data. Linear search is preferable when maintaining a sorted order is impractical.

Q5

Closest pair (real-time scenario)

A (2, 8)

B (12, 30)

C (40, 50)

D (5, 1)

E (12, 10)

F (3, 4)

Distance computation

Distance (A, F)

$= \sqrt{((3-2)^2 + (4-3)^2)}$

$= \sqrt{2}$

$= 1.41$

Nearest centry

A (2, 3)

F (3, 4)

minimum distance = 1.41

Total comparisons.

$\frac{6(6-1)}{2} = 15$

comparisons = 15.

Time complexity

Best = $O(n^2)$

Average = $O(n^2)$

Worst = $O(n^2)$